Hybrid BERT-CNN Architecture for Misinformation Detection

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Abstract

We combine BERT contextual embeddings with a convolutional classifier to detect misinformation. The hybrid model reaches 96.5 percent accuracy on an English benchmark.

Keywords: misinformation detection, BERT, convolutional neural network, hybrid model

1. Introduction

Pre-trained language models and convolutional networks capture different signals. We ask whether a hybrid improves misinformation detection over either alone.

2. Related Work

BERT classifiers are strong but costly, as surveyed in A Survey of Fake News Detection Using Machine Learning. CNN classifiers are cheap and simpler, as shown in Convolutional Neural Networks for Fake News Classification.

3. Methodology

BERT produces token embeddings that are fed into a CNN with several filter sizes. The model is trained end to end on an English news benchmark.

4. Results

The hybrid reached 96.5 percent accuracy, ahead of BERT alone at 95.7 percent and the CNN alone at 93.0 percent.

5. Limitations

The model lacks real-time evaluation. It has a high computational cost and was tested on English only.

6. Conclusion

A BERT-CNN hybrid gives the best accuracy among the compared models.

7. Future Work

Future work should build lightweight versions of the hybrid and evaluate it in real time.

References

[1] A Survey of Fake News Detection Using Machine Learning, Journal of Demonstration Studies, 2021.
[2] Convolutional Neural Networks for Fake News Classification, Journal of Demonstration Studies, 2019.

Note: this is a SYNTHETIC demonstration paper with fictional authors and invented results, created to demo ResearchGraph. It is not a real publication.